FILE PERMISSION AUDITOR

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Abstract

File Permission Auditor is a Linux-based C application that scans files, checks permissions, and identifies unsafe access. It helps users correct risky permissions, maintain audit logs, and generate security reports using Linux system calls and file handling.

Problem Statement & Objectives

Problem: Incorrect file permissions in Linux can allow unauthorized access or modification of files. Manually checking multiple directories is time-consuming.

Objective 1: To scan directories recursively and classify files as Safe, Warning, or Critical.

Objective 2: To correct unsafe permissions, maintain audit logs, and generate reports.

Scope: Designed for Linux users to audit file permissions using C, covering directory traversal, file handling, ownership, and access control.

Existing vs Proposed System

Aspect	Basic / Existing Approach	Proposed Project
Permission Checking	Manual checking of file permissions	Automatically checks file permissions using lstat()
Directory Scanning	Files checked individually	Recursively scans directories and subdirectories
Risk Identification	Difficult to identify unsafe permissions manually	Classifies files as Safe, Warning, or Critical
Permission Correction	Permissions changed manually	Allows users to correct permissions using chmod()
Audit & Reporting	No centralized record of changes	Maintains an audit log and generates a scan report

Feasibility Study

Technical: Uses C programming, Linux/Ubuntu, GCC, and system calls such as opendir(), readdir(), lstat(), and chmod().

Economic: Uses free, open-source tools. No additional hardware or hosting costs are required.

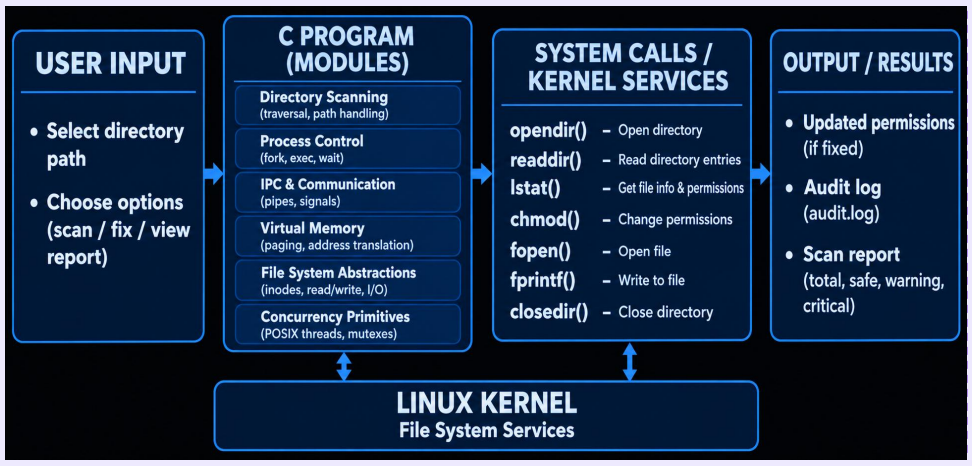
Operational: Simple menu-based interface for scanning directories, checking permissions, fixing unsafe permissions, and viewing reports. Requires suitable file access permissions.

Schedule: Completed within the semester timeline through planning, coding, testing, debugging, and final implementation.

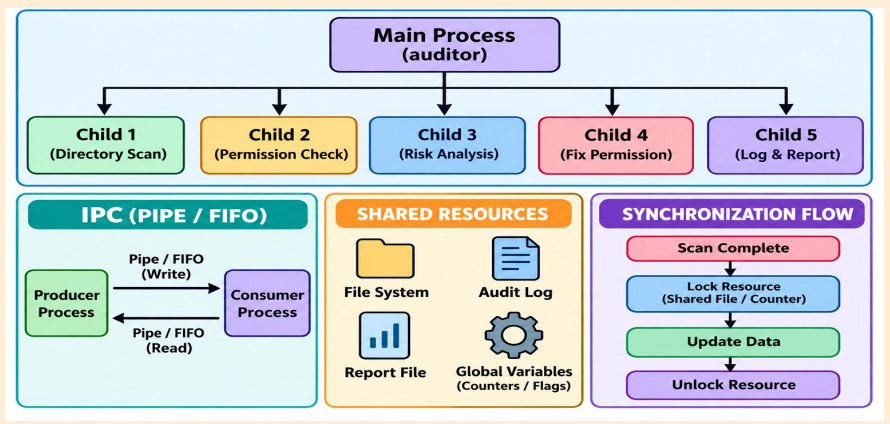
Tools & Technologies

- Programming: C
- Platform: Linux / Ubuntu (WSL)
- OS Concepts: File Systems, File Permissions, File Ownership, Directory Traversal, Access ControlSystem Calls / APIs: opendir(), readdir(), lstat(), chmod(), closedir()
- Tools: GCC, VS Code, Git & GitHub

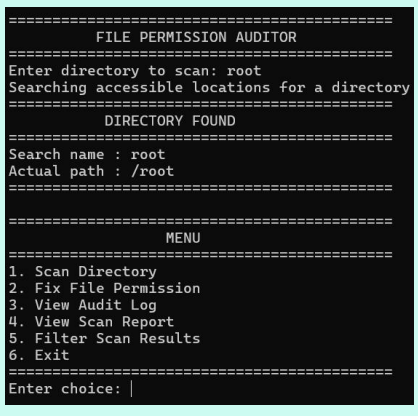
System Workflow / Architecture

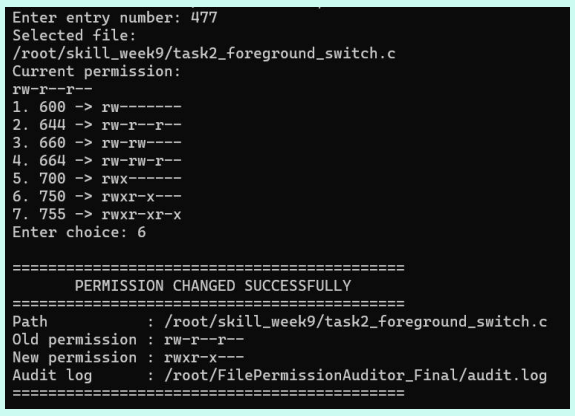


Process, IPC & Resource Design



Implementation Screenshots





Results & Testing

Sample Output: Displays file permissions and classifies files as Safe, Warning, or Critical.

Key Observations: Identifies unsafe permissions and records corrections in audit.log.

Testing: Tested normal cases, empty directories, invalid paths, and permission errors.

Conclusion

Implementation: Developed a C-based File Permission Auditor to scan directories, detect unsafe permissions, fix them, and generate audit logs and reports.

Objectives & OSSP Concepts: Meets the project objectives while demonstrating Linux file systems, file permissions, directory traversal, system calls, and file handling.

Future Scope

Enhancement 1: Add user and group ownership checks with more advanced permission analysis.

Enhancement 2: Improve scanning speed and add more detailed reports with better error handling.

References: Linux Man Pages, Operating Systems Textbook/Course Material, and GCC (GNU Compiler Collection) Documentation.